AI.302

Cyber-Physical Systems

COURSE OVERVIEW

Integrating computation, networking, and physical processes: CPS architecture, real-time coordination, and safety.

1

SESSION

60 MINUTES

ORIENTATION: THE CYBER-PHYSICAL MERGE

KEY TOPICS

- What is CPS
- Computation + networking + physics
- CPS vs. IoT
- Applications

TEACHING FOCUS

Blend the digital with the physical.

KEY TAKEAWAY CPS are systems where software moves matter.

CYBER-PHYSICAL SYSTEMS

2

SESSION

60 MINUTES

CPS ARCHITECTURE

KEY TOPICS

- Sensors, controllers, actuators
- Networked control
- Real-time operating systems
- System-of-systems

TEACHING FOCUS

Architect tight cyber–physical loops.

KEY TAKEAWAY Architecture binds computation to physics.

3

SESSION

60 MINUTES

REAL-TIME COORDINATION

KEY TOPICS

- Time-triggered vs. event-triggered
- Synchronization
- Scheduling
- Determinism

TEACHING FOCUS

Coordinate actions across time.

KEY TAKEAWAY Time is the glue of cyber–physical systems.

CYBER-PHYSICAL SYSTEMS

4

SESSION

60 MINUTES

SAFETY & RELIABILITY

KEY TOPICS

- Fault tolerance
- Redundancy
- Fail-operational design
- Safety standards

TEACHING FOCUS

Keep physical processes safe.

KEY TAKEAWAY Physical stakes demand high reliability.

5

SESSION

60 MINUTES

CPS IN ROBOTICS

KEY TOPICS

- Robotic CPS examples
- Smart manufacturing
- Autonomous vehicles
- Medical robotics

TEACHING FOCUS

Apply CPS to embodied systems.

KEY TAKEAWAY Modern robots are quintessential CPS.

CYBER-PHYSICAL SYSTEMS

6

SESSION

60 MINUTES

CPS DESIGN & REVIEW

KEY TOPICS

- CPS prototype
- Course recap and assessment
- Toward swarm robotics (AI.303)
- System validation

TEACHING FOCUS

Design a cyber–physical system end-to-end.

KEY TAKEAWAY CPS must be safe by design.